CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Six Semester of B. Tech. (EE) Examination

December 2013

EE309 Electric Power Utilization & Traction

Date: 06.12.2013, Friday **Time: 01:30 a.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Explain types of electric drives & write advantages of Electric drives. **[07]**

Q - 2 (a) Write condition necessary to achieve electric regenerative braking. and Explain regenerative braking for D.C. Shunt motor. **[07]**

(b) Explain Rheostatic braking for three phase induction motor. **[07]**

OR

Q - 2 Write Short notes: (Any Two) **[14]**

(1) mechanics of train movement

(2) Tramway and trolley bus

(3) Rotor side Speed control of 3-phas induction motor.

Q - 3 (a) Explain tractive effort for propulsion of train. **[07]**

(b) An electric train accelerates Uniformly from rest to a speed of 48 Km/hour in 24 seconds. Then coasting for 69 seconds against a constant resistance of 58 N/tone. & uniform braking to rest at 3.3 Km/hour/second in 11 seconds. Calculate (i) the acceleration (ii) coasting retardation and (iii) the schedule speed. If the station stops are 20 seconds duration, what would be the effect on schedule speed of reducing the station stops to 15 seconds duration, other condition remaining same, Allow 10 % for rotational inertia? **[07]**

OR

Q - 3 (a) Draw typical speed time curve for sub-urban service and Calculate Vm by trapezoidal speed time curve. **[07]**

(b) An electric train weighing 200 tonne has eight motors geared to driving wheels, each wheel is 90 cm diameter. Determine the torque developed by each motor to accelerate **[07]**

train to a speed of 48 kmph. in 30 seconds up gradient of 1 in 200. The tractive resistance is 50 newtons per tonne, the effect of rotational inertia is 10% of the train weight, the gear ratio is 4 to1 and gearing efficiency is 80 percent.

SECTION – II

Q - 4 (a) State different types of heating methods and write advantages of electric heating. [07]

Q - 5 (a) Explain basic types of reflection and Explain street Lighting. [06]

(b) Explain polar curves. [04]

(c) Write a short note on Butt Welding. [04]

OR

Q - 5 (a) Write the Comparison between tungsten Filament lamps and Fluorescent tubes. [06]

(b) Explain inverse squares law of illumination. [04]

(c) Explain measurement of candle power. [04]

Q - 6 (a) Derive flywheel calculation equation when motor load is increasing and motor load is decreasing. [07]

(b) A 25 hp., 3-phase, 10 pole, 50Hz induction motor provided with a flywheel has to supply a load torque of 800 N-m for 10seconds followed by a no- load period during which the flywheel regains the full speed. The full load slip of the motor is 4% and the torque- speed curve may be assumed linear over working range. Find the moment of inertia of the flywheel if the motor torque is not to exceed twice the full load torque. [07]

OR

Q - 6 (a) Explain electro-deposition process and discuss various factor governing the electro-deposition process [07]

(b) Explain any two modes of Heat transfer. [03]

(c) Give the comparison between AC and DC welding. [04]

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